Started on	Tuesday, 10 January 2023, 5:25 PM
State	Finished
Completed on	Tuesday, 10 January 2023, 5:26 PM
Time taken	40 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

What is the advantage of libc buffering?

Select one:

- a. it allows us to write more bytes using printf
- b. it reduces the number of context switches between user space and kernel space ✔
- c. it makes printf faster because it performs one `write()` syscall per call to `printf()`
- d. it doesn't require synchronizing the disk after every `write()`

libc buffering reduces the number of context switches between user space and kernel space. For this, it copies the string passed to printf() to an internal buffer inside libc. This buffer will be written at a given time (like when the buffer is full, for example).

The correct answer is: it reduces the number of context switches between user space and kernel space

Question **2**

Correct

Mark 2.00 out of 2.00

When is `read` blocking?

Select one:

- a. never
- b. when the buffer is empty ✔
- c. when the buffer is full
- d. any time we try to read something

`read` blocks when its associated buffer is empty because it doesn't have anything available to read.

The correct answer is: when the buffer is empty

Question **3**

Correct

Mark 2.00 out of 2.00

Which of the following is an advantage of using `dup2` instead of `dup`?

Select one:

- a. `dup2` allows duplicating a file descriptor into a file descriptor which doesn't have to be sequentially ascending to existing descriptors ✔
- b. `dup2` is executed in user space
- c. `dup2` is faster
- d. `dup2` performs slow operations in user space and fast operations in kernel space

The `dup2` call makes the file descriptor called newfd refer to the same open file description as the descriptor oldfd. It allows duplicating on a position that is not sequentially ascending to existing descriptors.

The correct answer is: `dup2` allows duplicating a file descriptor into a file descriptor which doesn't have to be sequentially ascending to existing descriptors

Question **4**

Correct

Mark 2.00 out of 2.00

What is an advantage of zero-copy?

Select one:

- a. it performs a single context switch ✔
- b. it doesn't perform any context switching
- c. data is copied multiple times, so we have redundancy
- d. data is copied into the app's internal buffer, so we have caching

The correct answer is: it performs a single context switch

Question **5**

Correct

Mark 2.00 out of 2.00

Which file descriptor is associated by default to `stdin`?

Select one:

- a. It differs from process to process
- b. 0 ✔
- c. 2
- d. 1

stdin has file descriptor 0 associated by default. stdout has fd 1, and stderr has fd 2.

The correct answer is: 0

◄ Curs 10: I/O (partea a 3-a): Recap

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Curs 12: Interacțiunea între aplicații (partea a 2-a) ►